Introduction to Inverse Problems and Imaging - 02624

Homework 2

1. Consider the 1D heat problem on $(0, \pi)$ for $T = 0.1$ modeled by the operator K. Take $x(s) = \sin(4s)$ and $y(t) = Kx(t)$. Let $N = 128$, define equidistant grids of s, t, and define by $Ax = y$ the discrete approximation of $Kx(t) = y(t)$. Consider the inverse problem

$$Ax = y$$

in the Bayesian setting with $Y = Ax + E$. We assume the likelihood and prior

$$X \sim \mathcal{N}(0, \gamma^2 I),$$
$$E \sim \mathcal{N}(0, \sigma^2 I),$$

with $\gamma = 1, \sigma = 10^{-2}$. Define the observation $y_{\text{obs}} = y + e$, where e is a realization of E.

Bayes formula gives

$$\pi_{\text{post}}(x) = \pi(x|y_{\text{obs}}) = \frac{\pi(y_{\text{obs}}|x)\pi_{\text{pr}}(x)}{\pi(y)} \propto \exp\left(-\frac{1}{2}(x - \overline{x})^T \Gamma_{\text{post}}^{-1}(x - \overline{x})\right),$$

with

$$\Gamma_{\text{post}} = \gamma^2 I - \gamma^4 A^T(\gamma^2 AA^T + \sigma^2)^{-1}A,$$
$$\overline{x} = \gamma^2 A^T(\gamma^2 AA^T + \sigma^2)^{-1} y_{\text{obs}}.$$

   (a) Derive $\overline{x} = R_\alpha y_{\text{obs}}$ (the ordinary Tikhonov) and show that the posterior covariance satisfies $\Gamma_{\text{post}} < \gamma^2 I$ in the sense of positive definite matrices.

   (b) Draw, say 100, samples from π_{post}. Compute the sample mean, sample variance, and plot the sample mean, posterior mean, and the posterior mean with 95% credibility intervals. Comment on the results.

   (c) Repeat the above numerical experiment in the situation where we sample the terminal condition only on the interval $(\pi/4, 3\pi/4)$. Comment on the results.
   (Hint: From your A above, take out a submatrix corresponding to the new observation interval.)

2. Consider the Abel problem for $x(s) = s^{3/4} - 1$. Discretize K by a matrix $A \in \mathbb{R}^{1024\times 1024}$ and discretize similarly $x(s)$ and $y(t) = Kx(t)$ by $x, y \in \mathbb{R}^{1024}$. Add random noise of magnitude 10% to y to get y^δ. Denote by D a finite difference matrix, so that Dx approximates $x'(s)$ (remember to scale with N).

   (a) Show that $x \in H^1(0, 1)$. Thus it makes sense to apply H^1 regularization in the inverse problem.

(b) We approximate the H^1-norm by $\|x\|_2^2 + \|Dx\|_2^2$.
Show that the problem

$$x_\alpha = \arg\min_x \|Ax - y^\delta\|_2^2 + \alpha(\|x\|_2^2 + \|Dx\|_2^2) \tag{1}$$

has a unique solution for any $\alpha > 0$.

(c) Find the solution to the ordinary Tikhonov problem (without D) with α obtained by the discrepancy principle. Show via a plot of the error ($\|x_\alpha - x\|$) how α relates to the optimal value.

(d) Find the solution to the problem (1), again choosing α by the discrepancy principle. Compare this solution to the previous without D.

Important notes:

- Deadline for the HW2 is November 19 at noon.
- All homeworks must be approved in order to qualify for the exam.
- Your background is different, and therefore you can choose to put emphasis differently in your solution.
- Demonstrate in your solution your understanding of the material and computational competencies.
- Each student should hand in a single pdf via DTU Learn. The number of pages for the main report must not exceed 7. Code can be put in an appendix.
- Comment on all results: What do you see, and why is it reasonable (or unreasonable)?
- Make quality plots with proper names on axis, legends and captions.